ETL_Traffic_Data_Analysis_Ipsit_Mohanty

California Traffic Collision Analysis (SWITRS) – ETL Assignment ETL/01 | Ipsit Mohanty

1. Problem statement and approach

Traffic collisions are a major public-safety risk, and agencies need evidence on where, when and under what conditions serious crashes happen. This report analyses 935,791 California collision records (2001–2021) from the Statewide Integrated Traffic Records System (SWITRS) with PySpark, to identify high-risk locations and times, the role of weather, road surface and lighting, and to recommend practical safety measures.

Approach. (1) Load the collisions, parties and victims tables and check the schema; (2) clean the data (sparse columns, data types, missing values, duplicates, outliers); (3) encode and reorder the cleaned data and save it; (4) explore it with univariate, bivariate and time-series analysis in Pandas, Matplotlib and Seaborn; (5) answer seven business questions with Spark SQL; (6) turn the findings into recommendations.

2. Data preparation and cleaning

The collisions table has 935,791 rows and 65 columns; after cleaning, 58 columns remain (63 with the five label-encoded columns added). The main data-quality issues and how each was handled are below.

Issue found	Evidence	Treatment
Highly sparse columns	7 columns are more than 60% missing (e.g. side_of_highway 72.9%, postmile 69.3%, pcf_violation_subsection 64.3%)	Dropped: caltrans_county, caltrans_district, state_route, postmile, location_type, side_of_highway, pcf_violation_subsection
Missing coordinates	latitude and longitude are 71.5% missing, but 266,742 records have valid coordinates	**Kept**, because they are still enough for a geographic scatter plot; plotted only where present
Wrong data types	Count columns stored as double	killed_victims, injured_victims, party_count and four other count columns cast to integer
Missing categories	e.g. 4,626 records (0.5%) have no weather recorded	Categorical columns (weather, road surface, road condition, lighting, collision factor, collision type) filled with 'Unknown'
Missing counts	Count columns with nulls	Filled with 0 (no recorded victims or parties)
Duplicates	Full-row duplicate check	0 duplicate rows found and removed
Dates and times	collision_date and collision_time stored as text/timestamp	Converted to datetime; year, month, hour and weekday derived
Very large case_id values	Some 2001 ids reach about 9.8E18, beyond a 64-bit integer	case_id left numeric (double) and used only as a join key

Outliers. The IQR method flags 33,313 records for injured_victims, 298,314 for party_count and none for killed_victims. These are not errors. party_count is discrete with Q1 = Q3 = 2, so any crash that is not a two-party crash is flagged; crashes with three or more injured victims are rare but real multi-vehicle incidents. As the assignment allows, **no outlier was removed**; medians and counts are used for summaries.

3. Exploratory data analysis

Variable types. Categorical: collision_severity, weather_1, road_surface, lighting, type_of_collision. Numerical: injured_victims, killed_victims, party_count (plus latitude, longitude and the derived year, month, hour).

3.1 Collision severity and victims

Most collisions are minor: **60.96% are property-damage-only** (570,480), 24.20% involve complaint of pain (226,492), 12.47% other visible injury (116,681) and **2.37% severe injury** (22,138). The sample contains no

fatal collisions.

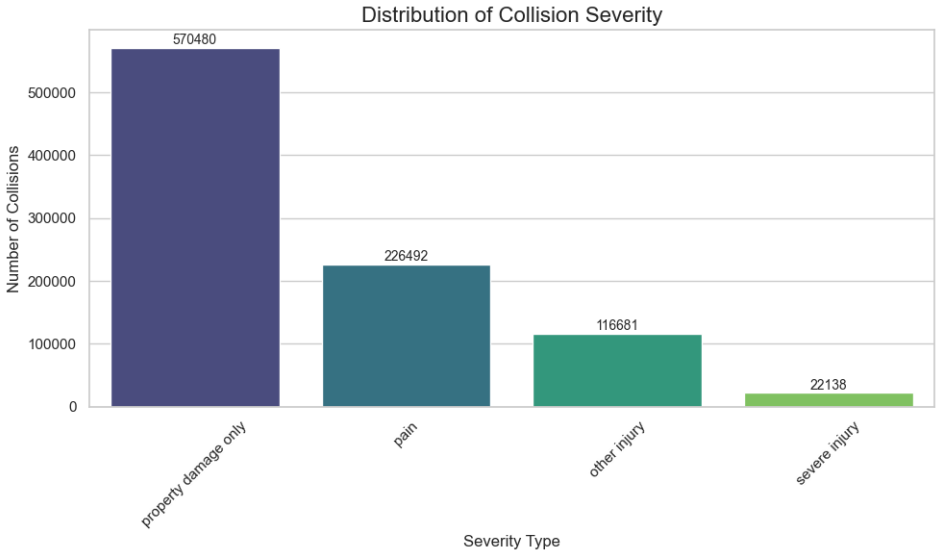


Figure 1. Distribution of collision severity

The victims table gives a victim-age profile (925,096 valid ages after removing 0, the 998/999 unknown codes and values above 100): median age 27, mean 31.3, most common age 18, interquartile range 17–45. About 44% of victims are aged 16–35, so young adults carry most of the injury burden.

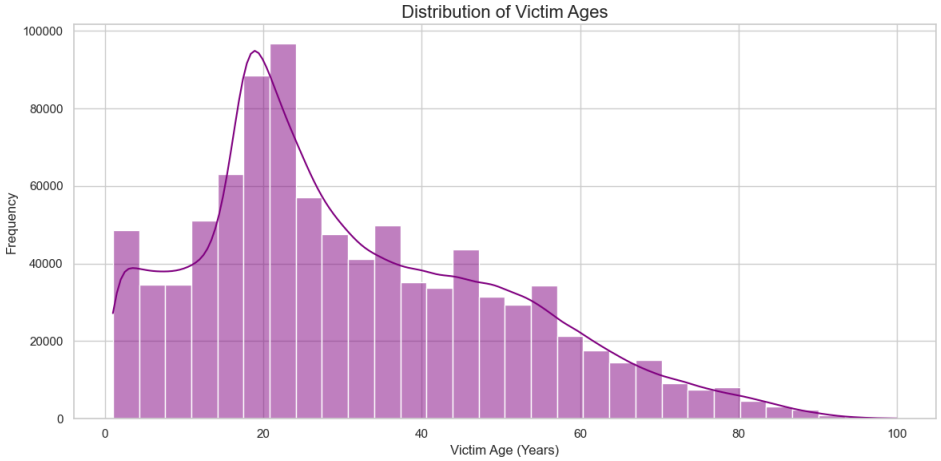


Figure 2. Distribution of victim ages

Severity and number of victims. Using the injured-victim count recorded on each collision, severe-injury crashes average 1.62 injured people, other-injury crashes 1.51, pain crashes 1.37 and property-damage-only crashes none by definition. Severity therefore rises with the number of people hurt, although the gap between the injury categories is modest.

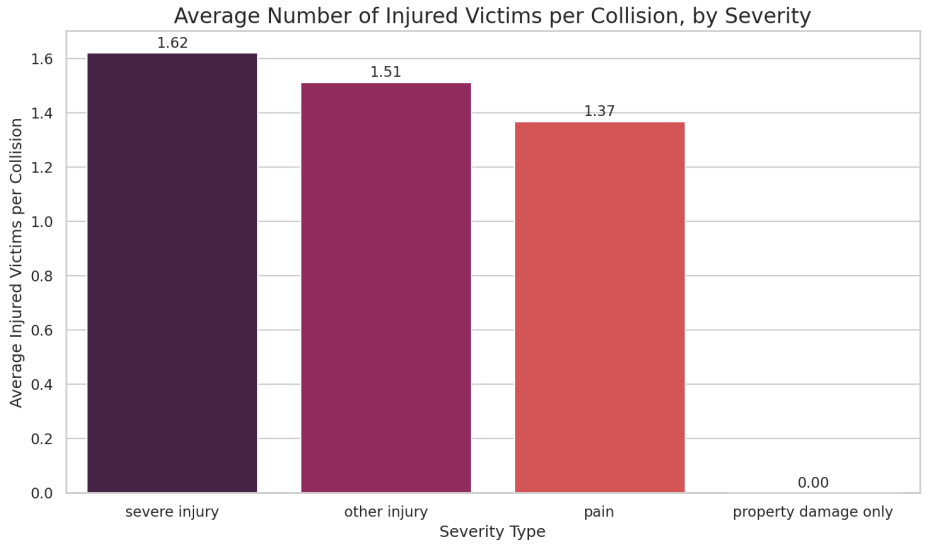


Figure 3. Average injured victims per collision by severity

Note: the sample victims table links to only 9.3% of the sampled collisions (about 87,000 of 935,791, from a PySpark inner join on case_id), so counting rows from the victims table would understate victims per crash and mainly reflect how many collisions happen to have matching rows. The injured_victims field is complete and is used instead.

3.2 Weather, road surface and lighting

Weather. 82.3% of collisions (769,927) happen in clear weather, followed by cloudy and rainy conditions. Raw counts mostly reflect how often each condition occurs, so the severity *mix* within each condition is more informative:

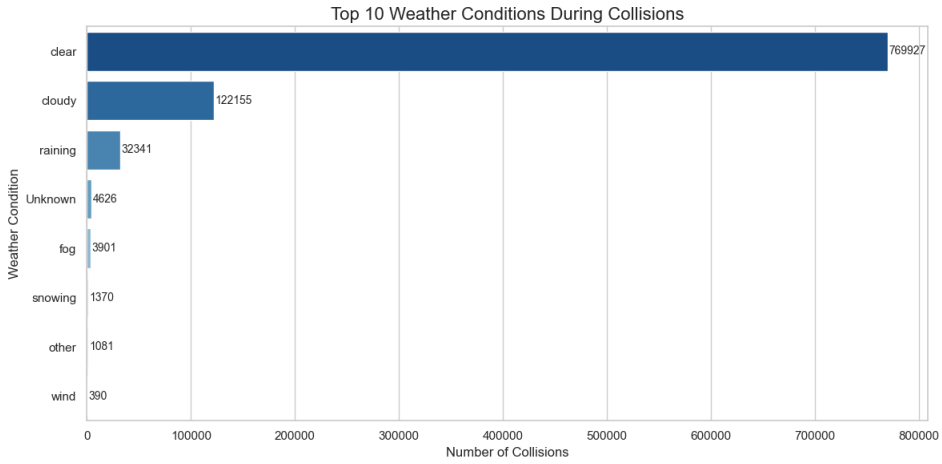


Figure 4. Collisions by weather condition

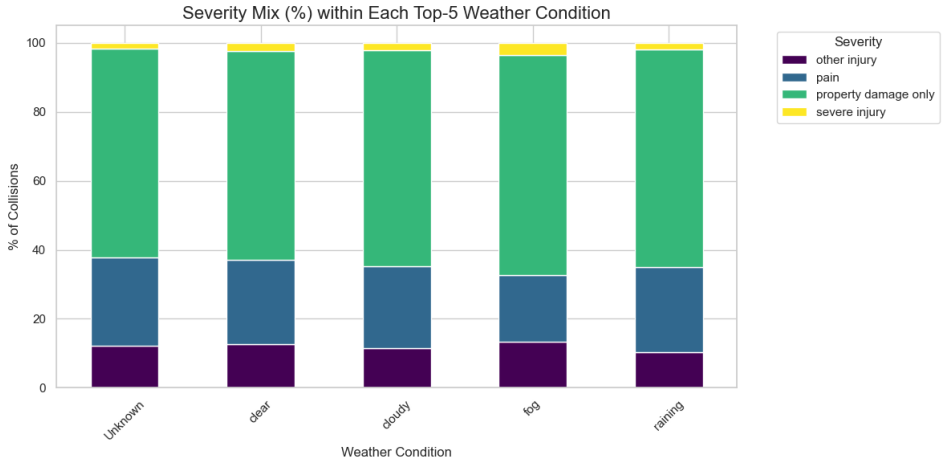


Figure 5. Severity mix (%) within each weather condition

Weather	Severe-injury share
Fog	3.59%
Clear	2.42%
Cloudy	2.11%
Raining	1.92%
Unknown	1.66%

Fog is the most severe condition and rain the mildest, perhaps because drivers slow down in rain. Fog has far fewer crashes than clear weather, so its percentage should be read as indicative, not proven.

Lighting. Daylight accounts for 625,574 collisions (66.8%), dark roads with street lights 195,867 (20.9%) and dark roads without street lights 74,634 (8.0%). Severity tells a different story:

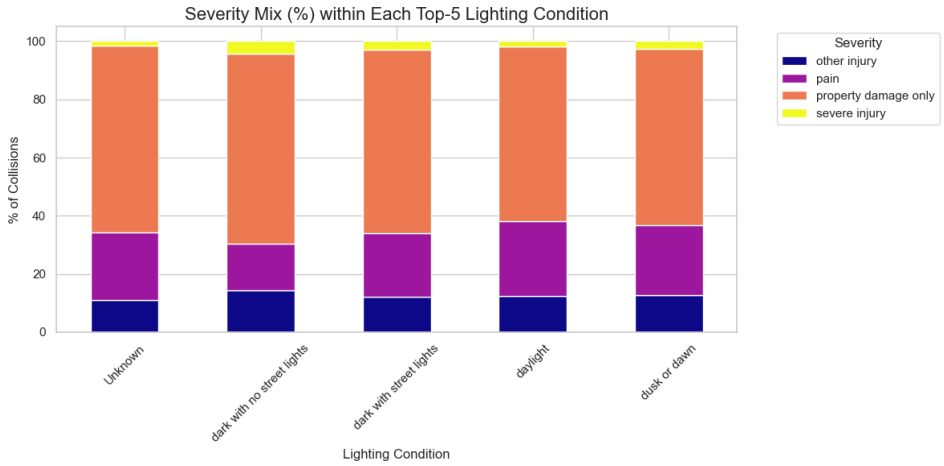


Figure 6. Severity mix (%) within each lighting condition

Lighting	Severe-injury share
Dark, no street lights	4.35%
Dark, street lights	2.97%
Dusk or dawn	2.74%
Daylight	1.92%
Unknown	1.40%

A collision on an unlit road is about 2.3 times as likely to cause a severe injury as one in daylight, even though daylight has by far the most collisions. This is the clearest environmental risk signal in the data.

Road surface. 90.4% of collisions (845,637) occur on dry roads and 8.2% (76,833) on wet roads, so most crashes are not weather-driven.

3.3 When and where collisions happen

Day of week. Collisions are highest on Friday (154,544) and lowest on Sunday (109,081), with Monday to Thursday close together at 131,000–139,000.

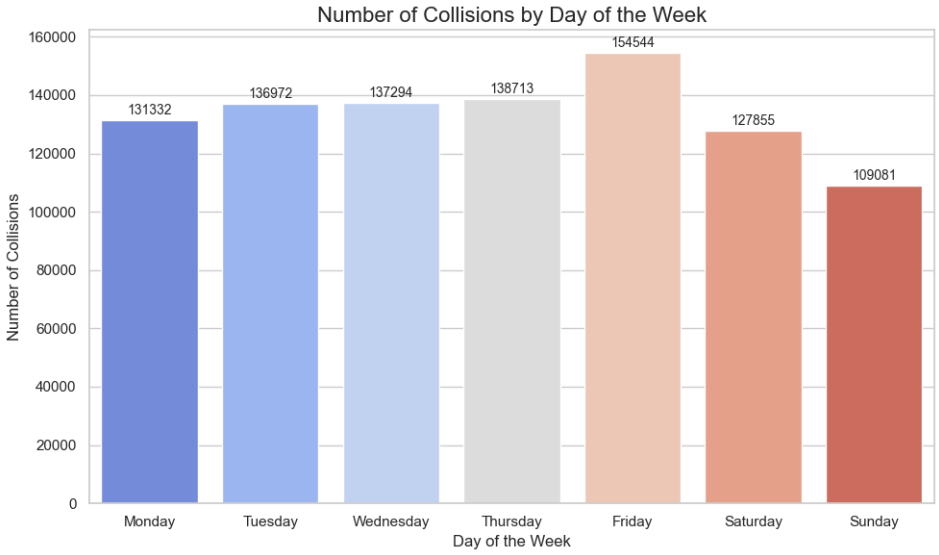


Figure 7. Collisions by day of week

County. Los Angeles County alone has 284,100 collisions (30.4% of the total); the top five counties together account for 55.0%.

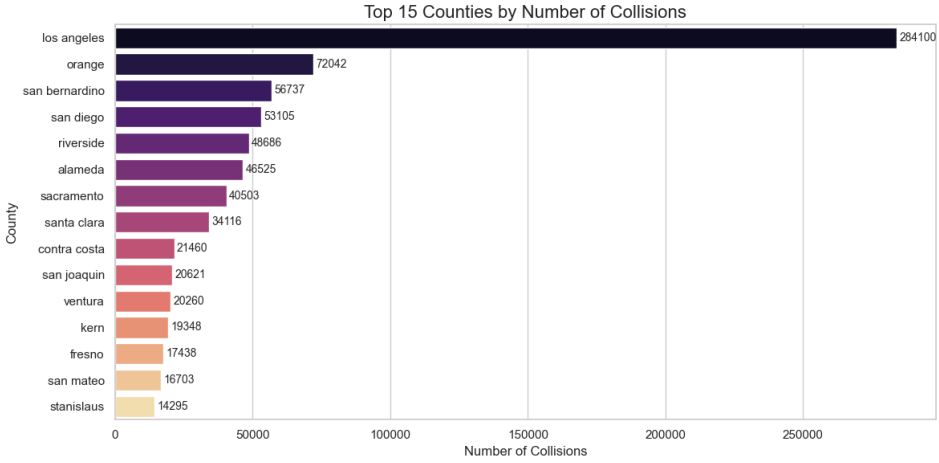


Figure 8. Collisions by county

Geographic scatter. 266,742 collisions (28.5%) have usable coordinates. They cluster around Los Angeles, the Bay Area, Sacramento and San Diego and trace the major highway corridors. Because 71.5% of records lack coordinates, the map shows where located collisions cluster and may under-represent some areas.

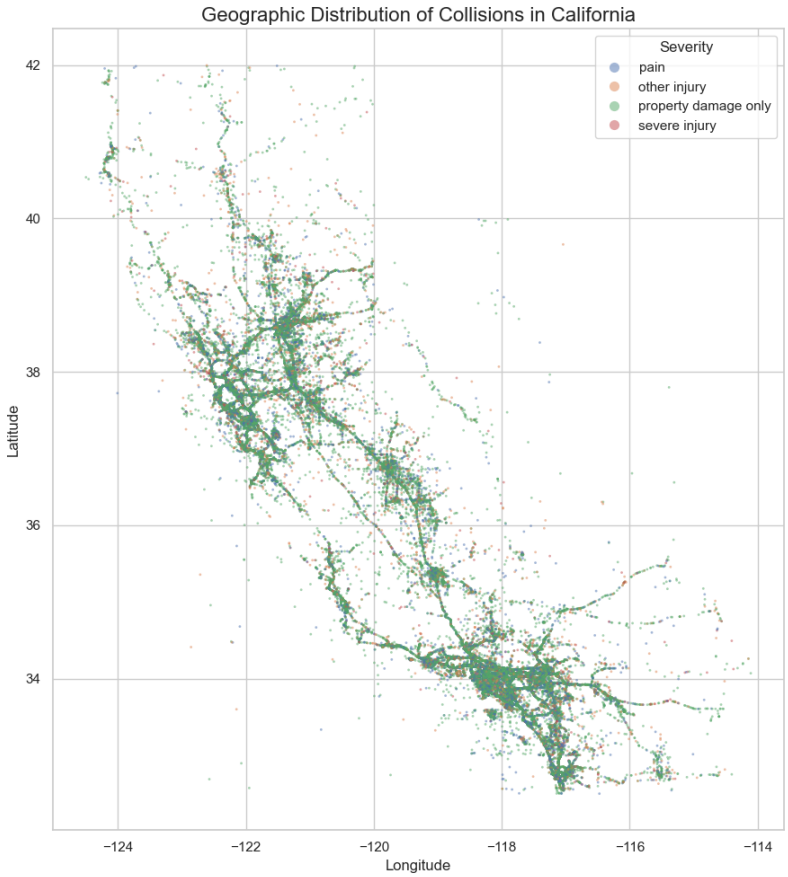


Figure 9. Geographic distribution of collisions (60,000-point random sample, coloured by severity)

3.4 Trends over time

Yearly. Collisions were highest in 2002–2004 (about 53,700 a year), fell by about 29% to a low of 38,234 in 2013, rebounded to 49,167 in 2016 and then declined again. 2020 (36,020) is likely linked to reduced travel during the pandemic, although the data alone does not prove the cause. **2021 is a partial year** (13,198 records), so its drop is a coverage effect, not a real decline.

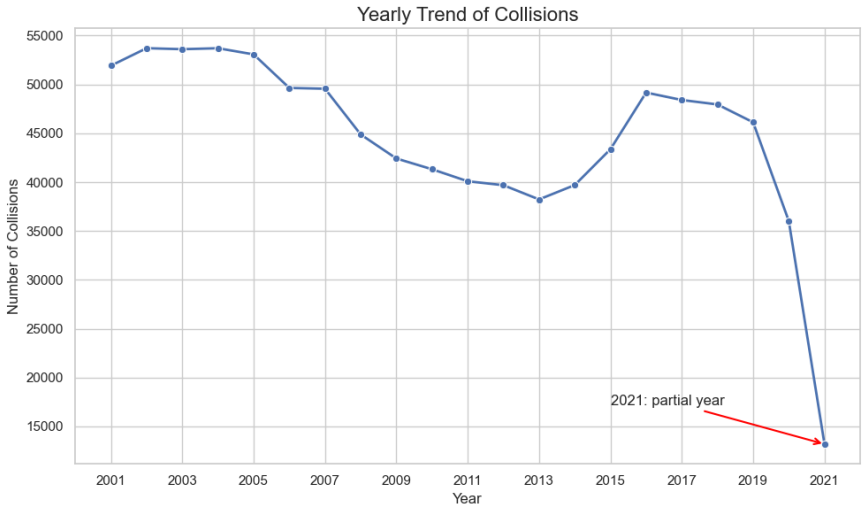


Figure 10. Yearly collision trend

Monthly. The pattern is fairly flat, from 74,762 in June to a peak of 83,274 in October, with March (82,167) a secondary high.

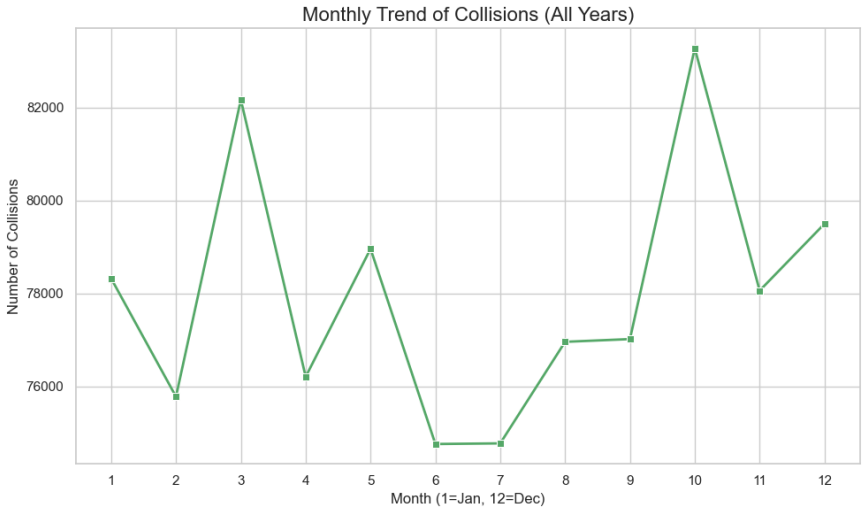


Figure 11. Monthly collision trend (all years)

Hourly. Collisions follow commuter traffic: a morning bump at 07:00–08:00 (about 47,000–49,000) and a larger afternoon peak, with **17:00 the busiest hour (73,255)**, followed by 15:00 (70,857) and 16:00 (67,432). The quietest hour is 04:00 (10,650).

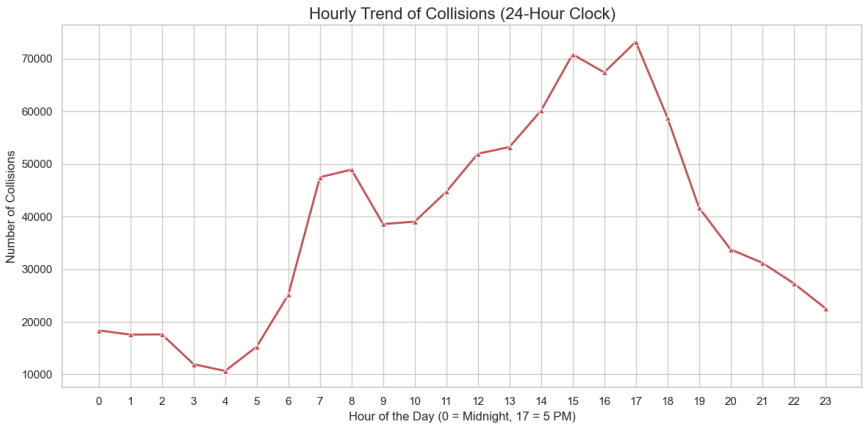


Figure 12. Hourly collision trend

4. Querying results (Spark SQL)

Question	Result
Top 5 counties by collisions	Los Angeles 284,100; Orange 72,042; San Bernardino 56,737; San Diego 53,105; Riverside 48,686
Month with most collisions	October (month 10): 83,274
Most common weather condition	Clear: 769,927 (82.3%)
Percentage of fatal collisions	**0 of 935,791 (0.00%)**: no fatal severity and killed_victims = 0 throughout the sample. Severe injuries are 22,138 (2.37%), reported separately
Most dangerous time of day	17:00 (5 PM): 73,255 collisions
Top 5 road surface conditions	Dry 845,637; wet 76,833; unknown 8,141; snowy 4,122; slippery 1,048
Top lighting conditions	Daylight 625,574; dark with street lights 195,867; dark with no street lights 74,634

5. Conclusions and recommendations

Key findings. Collisions are concentrated in space (Los Angeles County has 30%), in time (Friday, and the 3–6 PM commute with a peak at 5 PM) and in ordinary conditions (clear, dry, daylight). Severe injuries are rare (2.37%) but are disproportionately linked to darkness without street lights, fog and vulnerable road users.

1. High-risk locations and peak times

- Prioritise Los Angeles County, Orange, San Bernardino, San Diego and Riverside (55% of collisions) for infrastructure review.
- Concentrate enforcement and signal-timing work in the 3–6 PM window and on Fridays.

2. Traffic management and road design

- Rear-end crashes are the most common type (312,132, 33%), pointing to following distance and signal/queue management at busy junctions.
- Retime signals for the evening peak and use variable message signs for congestion ahead.

3. Pedestrians, cyclists and motorcyclists

- A supplementary check on the collision flags shows far higher severity for vulnerable road users: severe injuries make up 13.7% of pedestrian-involved, 16.4% of motorcycle-involved and 7.2% of bicycle-involved collisions, against 2.4% overall.
- Policy options: protected crossings and refuge islands, lower speed limits near high-pedestrian areas, protected bike lanes and motorcyclist-awareness campaigns.

4. High-risk zones for proactive intervention

- Use the geographic density map to flag corridors around Los Angeles, the Bay Area, Sacramento and San Diego; combine collision density with severity to rank locations.

5. Environmental factors

- **Lighting is the strongest lever:** 4.35% of crashes on unlit roads are severe versus 1.92% in daylight. Prioritise street lighting and reflective markings on unlit segments.
- Fog has the highest severe share among weather types (3.59%); fog warning systems and speed advisories are justified. Since 82% of crashes happen in clear weather and 90% on dry roads, campaigns should target speeding and distraction, not just bad weather.

6. Predictive models for hotspots

- Next step: train a classification or clustering model (for example gradient boosting on county, hour, weekday, lighting, weather and collision type, or spatial clustering of coordinates) to forecast hotspots, validated on a held-out period. Data quality must improve first: mandatory GPS coordinates and automated weather capture at dispatch would close the 71.5% location gap and the missing-weather records.

6. Assumptions and limitations

- The analysis uses the provided sample CSV files (collisions, parties, victims) rather than the full .sqlite database; results describe this sample.
- The sample has **no fatal collisions**, so the fatality rate is 0% and severity analysis focuses on severe injury.
- Missing categorical values were labelled 'Unknown' and missing counts set to 0; no rows were deleted. Sparse columns (>60% missing) were dropped except latitude/longitude.
- Outliers were identified with the IQR rule and retained, as they represent valid events.
- case_id was kept numeric because some 2001 ids exceed the 64-bit integer range; the sample victims table overlaps with only 9.3% of sampled collisions, so victim counts use the collisions table's injured_victims field.
- 2021 is a partial year and is not interpreted as a decline.
- The cleaned dataset was saved locally as Parquet and also exported as CSV for BI use; AWS S3 was not used. Spark's own Parquet writer needs Hadoop libraries that were not available on the local Windows machine, so a Pandas writer was used.

- Allowed libraries (NumPy, Pandas, Matplotlib, Seaborn, Plotly) were used for analysis; Spark handled loading, cleaning and SQL queries.

Optional visualisation step. The cleaned data was exported to CSV for use in Tableau. A live S3 connection with dynamic refresh was not set up.